

GROSS FIELD, CA, USA

WMO: 720406

Lat: 38.150N

Lon: 122.550W

Elev: 4

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00135

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.5	32.4	19.1	14.3	58.1	24.9	19.0	51.5	22.7	57.5	18.8	54.4	1.1	190	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.7	94.5	66.4	90.4	65.8	86.2	64.7	69.3	88.5	67.8	85.6	66.4	83.2	7.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	86.5	72.8	61.1	80.5	72.9	59.3	75.4	70.5	33.2	88.8	32.0	85.6	30.8	83.2	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.2	17.0	15.3	DB	25.3	101.5	2.6	3.5	23.5	104.1	22.0	106.2	20.5	108.1	18.7	110.7	(4)
(5)			WB	24.9	72.5	2.3	1.3	23.3	73.5	21.9	74.2	20.6	75.0	19.0	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	48.7	51.2	53.8	57.1	60.3	65.3	67.3	67.6	66.8	61.9	53.9	47.8	(6)	
	DBStd	8.52	4.37	4.83	4.59	5.08	4.74	5.19	4.07	3.97	4.84	4.71	5.05	5.31	(7)	
	HDD50	262	76	39	17	4	0	0	0	0	0	0	21	105	(8)	
	HDD65	2755	506	387	348	242	161	56	20	17	32	120	334	532	(9)	
	CDD50	3367	35	72	134	217	319	460	538	544	504	369	137	38	(10)	
	CDD65	387	0	0	0	6	16	65	93	97	86	24	0	0	(11)	
	CDH74	6664	1	10	25	203	438	1223	1409	1336	1407	595	16	1	(12)	
	CDH80	2755	0	0	1	55	164	550	574	553	665	193	0	0	(13)	
(14) Wind	WSAvg	5.1	4.3	5.3	5.4	6.3	6.5	5.9	5.6	5.0	4.9	4.2	3.9	4.3	(14)	
(15) Precipitation	PrecAvg	34.7	6.8	6.8	3.7	1.9	1.3	0.2	0.0	0.1	0.1	1.6	3.5	7.6	(15)	
	PrecMax	51.8	20.6	20.9	13.1	6.5	4.1	0.9	0.0	1.1	0.2	5.2	9.8	17.4	(16)	
	PrecMin	17.6	0.6	0.4	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.7	(17)	
	PrecStd	9.9	5.0	4.6	3.7	1.6	1.3	0.3	0.0	0.2	0.1	1.5	2.8	5.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.6	74.9	77.2	86.0	90.7	98.6	96.7	98.7	99.0	90.8	75.5	65.8	(19)	
		MCWB	51.5	58.2	60.7	64.5	65.3	69.4	66.5	67.5	65.6	62.4	59.3	48.7	(20)	
	2%	DB	63.6	69.5	72.6	80.7	84.3	91.2	90.7	91.1	93.0	85.5	72.6	62.9	(21)	
		MCWB	52.7	57.6	58.9	62.0	63.3	66.7	67.3	66.5	64.9	60.7	57.7	53.3	(22)	
	5%	DB	60.6	64.4	68.0	73.5	78.7	86.3	86.1	85.8	88.2	81.2	68.3	59.4	(23)	
		MCWB	52.8	54.4	56.4	59.5	61.3	64.8	66.4	65.8	64.3	59.7	56.8	53.3	(24)	
	10%	DB	57.2	61.4	64.1	68.3	72.8	81.3	81.9	81.5	81.8	76.5	64.2	57.2	(25)	
		MCWB	52.3	54.3	54.8	57.4	59.4	63.6	65.4	65.2	63.7	59.2	54.9	52.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.1	60.9	63.1	66.5	68.3	71.6	70.8	70.2	69.3	66.8	63.4	60.5	(27)	
		MCDB	61.7	69.2	73.9	81.3	86.1	93.2	91.0	90.7	90.0	79.2	72.7	61.9	(28)	
	2%	WB	57.0	59.0	60.3	63.2	64.8	68.9	68.8	68.3	67.3	64.5	61.3	58.3	(29)	
		MCDB	59.8	65.5	68.4	77.2	81.8	88.0	87.3	86.6	86.1	77.0	67.8	59.8	(30)	
	5%	WB	55.0	57.2	58.5	60.9	62.3	66.6	67.2	66.8	65.7	62.9	59.4	56.3	(31)	
		MCDB	57.9	63.3	65.1	72.5	76.2	83.4	84.1	83.8	83.4	74.3	64.5	57.6	(32)	
	10%	WB	53.6	55.1	56.7	58.4	59.9	64.2	65.4	65.3	64.1	61.1	57.2	54.4	(33)	
		MCDB	56.2	59.7	62.4	66.8	71.6	78.7	81.6	80.9	79.2	71.8	62.4	55.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	19.6	20.2	22.4	24.2	29.0	30.7	30.3	32.2	28.6	22.7	18.1	(35)	
		MCDBR	24.5	27.1	28.3	32.8	36.2	40.2	38.4	39.2	43.2	36.8	28.5	21.1	(36)	
		MCWBR	15.1	15.8	16.1	17.4	17.8	17.9	17.3	17.0	17.7	16.3	15.3	13.3	(37)	
	5% WB	MCDBR	16.6	22.3	23.2	29.4	32.0	35.9	35.5	34.6	35.9	29.6	21.9	11.7	(38)	
		MCWBR	12.6	14.2	14.3	16.6	17.2	17.7	17.0	16.2	16.3	14.9	13.5	9.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.320	0.324	0.334	0.354	0.359	0.355	0.361	0.341	0.341	0.325	0.318	0.321	(40)	
	taud		2.560	2.550	2.518	2.428	2.444	2.464	2.445	2.502	2.496	2.581	2.573	2.562	(41)	
	Ebn at Noon		275	288	295	293	292	292	289	294	288	284	273	266	(42)	
	Edh at Noon		24	27	31	36	36	35	36	33	31	26	23	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		738	1034	1461	1928	2255	2479	2377	2111	1774	1284	874	641	(44)	
	RadStd		112	135	142	158	144	111	81	82	67	89	61	101	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (45 neighbors)		N/A	N/A	-2.18	N/A	N/A	N/A	N/A	-194	+243	N/A

Nomenclature: See separate page